

Iram Anwar

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412-494-8746 | Crozet, VA

Education

University of Waterloo, Waterloo, Canada

Sept. 2022 - May 2027

Candidate for BASc in Mechatronics Engineering

Relevant Courses: DSA, RTOS, Dynamics, Sensors & Instrumentation, Control Systems, Electromech Machine Design

Lab Experience: Signal Processing, FPGA, Soldering, PLC (RSLogix, TwinCAT), STM32 (C), MATLAB, Stewart Platform

Skills

Design: SolidWorks, Fusion 360, Inventor, AutoCAD, GD&T, DFMA, Cura

Prototyping/Manufacturing: CNC, Laser Notching, Welding (Laser, Resistance), Winding, CT, X-ray

Software: C++, Python, C, ROS2, Linux, Git, Bash, OpenCV, VHDL, G&M, MATLAB, MiniTab

Hardware: NVIDIA Jetson, STM32, Arduino, Allen Bradley/Beckhoff PLCs, USART

Experience

Symbotic

May 2026 - Aug 2026

Robotics Hardware Test Engineering Intern | Boston, MA

- Determined battery endurance for Symbotic's [breakpack](#) products and justified a **\$2.5 mil increase** in product budget
- Instrumented a [Minibot](#) with a voltage divider & sense resistor for **coulombic counting**, estimating energy consumed

Whoop

Jan 2026 - May 2026

Battery Engineering Co-op | Boston, MA

- Created and executed Whoop's test plan for next-generation battery packs to meet PRD & Safety requirements
- Validated battery fuel gauge IC accuracy using battery cycling equipment and developer kits impacting current UX
- Built dashboards to analyze accelerated aging and capacity trends

Tesla

May 2025 - Aug 2025

Cell Manufacturing R&D Intern | Fremont, CA

- Developed new processes for Tesla's 4680 tabless cell program contributing to a **10% increase** in output of total cells
- Operated complex **winding machines** to conduct feasibility and validation trials on future tabless technologies
- Designed an **embedded system** in **C** using an **STM32** to characterize the thermal resistance of formed battery cells
- Gained hands-on and theoretical knowledge of battery cell manufacturing from electrode production to cell formation

Tesla

Aug 2024 - Dec 2024

Cell Manufacturing Engineering Intern | Austin, TX

- Led a **high priority manufacturing project** that contributed to a **70%** reduction in quality escapes on 4680 cells
- Incorporated fluid dynamics, controls, and mechanics into a tangible **poka-yoke closed loop exhaust flow upgrade**
- Performed root cause analyses, process validation testing, and DOEs on vertically integrated manufacturing machines
- Used SQL and dashboards investigating yield, **OEE**, availability, and primary responses to prove effectiveness of trials

MICRO

Jan 2024 - Apr 2024

Process Development/Automation Engineering Co-op | Somerset, NJ

- Created and tested processes used to create robot-operated minimally-invasive medical instruments (IQ, OQ, PQ)
- Conducted **DOEs** on an **Amada resistance welder**, a **heat shrink machine**, and an **IPG laser welding machine**
- Decreased cycle times by **20%** for laser welders, laser cutters, and CNC machines through **G&M code** optimization
- Utilized **GD&T** & **DFMA** in **SolidWorks** for fixtures used in heat shrinking and Instron tensile testing processes

University of Waterloo Mars Rover Team

Dec 2022 - Dec 2024

Mechanical/Software Team Member | Waterloo, ON

- Integrated **OpenCV** and **PyQT** to stream live video from cameras onto a driver **GUI** with **ROS2** pub/sub in **C++**
- Contributed to hardware functionality testing on **NVIDIA Jetson**, O-Drive motor controllers, DC motors, and **STM32s**